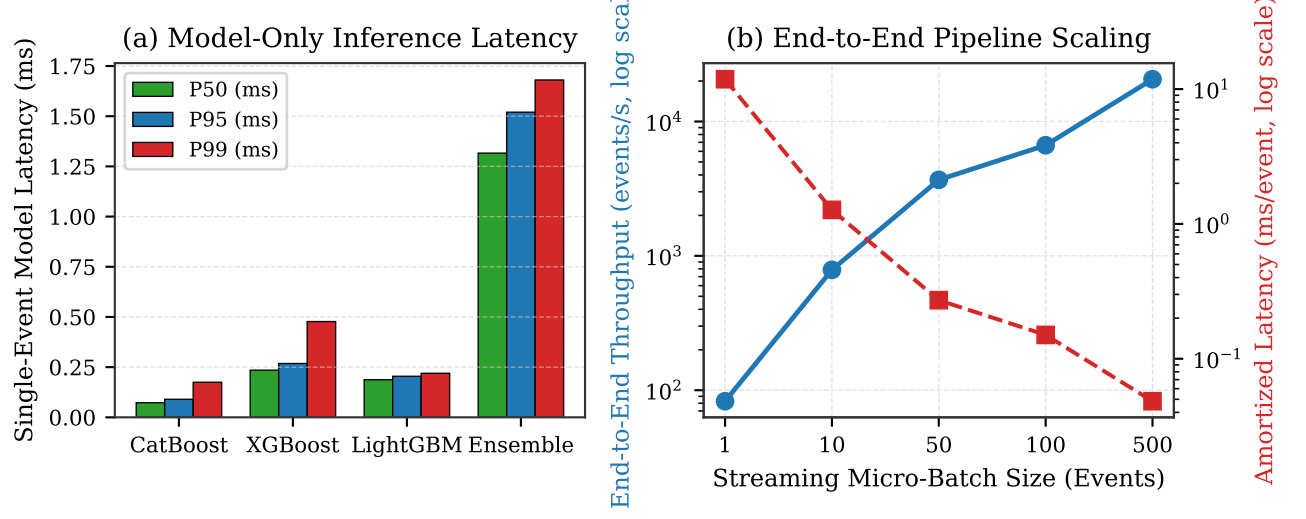
(a) Model-Only Inference Latency
Single-Event Model Latency (ms)
P50 (ms)
P95 (ms)
P99 (ms)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
CatBoost
XGBoost
LightGBM
Ensemble
(b) End-to-End Pipeline Scaling
End-to-End Throughput (events/s, log scale)
Amortized Latency (ms/event, log scale)
10^2
10^3
10^4
10^{-1}
10^0
10^1
1
10
50
100
500
Streaming Micro-Batch Size (Events)